



3 Complex Analysis 2024

Tutorial 8

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Liouville's theorem

Problem 32 Let f be an entire function. Suppose that $|f(z)| \geq 1$ for all $z \in \mathbb{C}$. Show that f is constant.

Solution.

Consider the function $\frac{1}{f(z)}$, $z \in \mathbb{C}$. Since $f(z) \neq 0$ for $z \in \mathbb{C}$, the function $\frac{1}{f(z)}$ is holomorphic in $\mathbb{C}$, i.e., it is an entire function. By assumption, $\left|\frac{1}{f(z)}\right| \leq 1$ for all $z \in \mathbb{C}$, i.e. $\frac{1}{f(z)}$ is bounded on $\mathbb{C}$. By Liouville's Theorem, $\frac{1}{f}$ is a constant function. It follows that also f is a constant function.

Identity theorem

Problem 33

- (a) Is there a function f that is holomorphic at 0 and satisfies $f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$?
- (b) Is there a function f that is holomorphic at 0 and satisfies $f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{n^3}$ for all $n \in \mathbb{N}$?

Solution.

First notice that f being holomorphic at 0 means that f is holomorphic in a certain neighborhood D of 0. The points $\frac{1}{n}$ will lie in this neighborhood at least for $n \geq n_0$ with some $n_0 \in \mathbb{N}$.

- (a) Yes, the function $f(z) = z^2$ satisfies the assumptions.
- (b) There is no such function. Since $f(z)$ coincides with z^3 on the set $\left\{\frac{1}{n} : n \geq n_0\right\}$ which has a limit point $0 \in D$, we must have $f(z) = z^3$ in D . On the other hand, $f(z)$ coincides with $-z^3$ on the set $\left\{-\frac{1}{n} : n \geq n_0\right\}$ which has a limit point $0 \in D$, which implies $f(z) = -z^3$ in D . This is a contradiction.

Gauss' mean value theorem

Problem 34 Prove Gauss' Mean Value Theorem:

Let f be a holomorphic function in the open disc $D(z; R)$ with some $z \in \mathbb{C}$, $R > 0$. Let $0 < r < R$. Then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta.$$

Solution.

We apply Cauchy's Integral Formula with the circle $\gamma(\theta) = z + re^{i\theta}$, $\theta \in [0, 2\pi]$. We have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dz = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z + re^{i\theta})}{re^{i\theta}} rie^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta.$$

Problem 35 Evaluate $\int_0^{2\pi} \sin^2 \left(\frac{\pi}{6} + 2e^{it} \right) dt$.

Solution.

The function $z \mapsto \sin^2 z$ is holomorphic in $\mathbb{C}$. By Gauss' Mean Value Theorem,

$$\int_0^{2\pi} \sin^2 \left(\frac{\pi}{6} + 2e^{it} \right) dt = 2\pi \sin^2 \left(\frac{\pi}{6} \right) = 2\pi \left(\frac{1}{2} \right)^2 = \frac{\pi}{2}.$$

Maximum modulus principle

Problem 36

(a) Let f be holomorphic in the disc $D(z_0; R)$ with some $z_0 \in \mathbb{C}$ and $R > 0$. If $|f|$ is constant in $D(z_0; R)$, then f is constant in $D(z_0; R)$. Hint: Use the Cauchy-Riemann Equations.

(b) Prove the Local Maximum Modulus Theorem:

Let f be holomorphic in the disc $D(z_0; R)$ with some $z_0 \in \mathbb{C}$ and $R > 0$. Assume that $|f(z)| \leq |f(z_0)|$ for all $z \in D(z_0; R)$. Then f is constant in $D(z_0; R)$.

(c) Prove the Maximum Modulus Principle:

Let G be a bounded region. Let f be holomorphic in G and continuous in $\overline{G}$. Then $|f|$ attains its maximum on the boundary of G which we denote by ∂G . If f is non-constant, then

$$|f(z)| < \max_{w \in \partial G} |f(w)| \quad \text{for any } z \in G.$$

Hint: Use the Local Maximum Modulus Theorem and the Identity Theorem.

Solution.

- (a) Let $f = u + iv$, and $|f(z)| = c$ for all $z \in D(z_0; R)$. If $c = 0$, then $f \equiv 0$ in $D(z_0; R)$ and we are done. So assume that $c \neq 0$. We have $u^2 + v^2 = c^2$ everywhere in $D(z_0; R)$. Differentiating this identity with respect to x and with respect to y , we obtain equations

$$uu_x + vv_x = 0, \quad uu_y + vv_y = 0.$$

The Cauchy-Riemann Equations say that $u_x = v_y$, $u_y = -v_x$. Eliminating v_x and v_y in the above equations, we obtain

$$uu_x - vu_y = 0, \quad uu_y + vu_x = 0. \quad (*)$$

Now multiply the first equation by u , the second equation by v and add them:

$$(u^2 + v^2)u_x = 0,$$

i.e.

$$c^2 u_x = 0.$$

Since $c \neq 0$, we obtain that $u_x = 0$ everywhere in $D(z_0; R)$. Then by the Cauchy-Riemann Equations $v_y = 0$ everywhere in $D(z_0; R)$.

Now multiply the first equation in $(*)$ by v , the second equation by u , and subtract the first equation from the second one:

$$(u^2 + v^2)u_y = 0,$$

i.e.

$$c^2 u_y = 0.$$

Since $c \neq 0$, we obtain that $u_y = 0$ everywhere in $D(z_0; R)$. Then by the Cauchy-Riemann Equations $v_x = 0$ everywhere in $D(z_0; R)$.

We deduce that f is constant.

- (b) Take r with $0 < r < R$. By Gauss' Mean Value Theorem,

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

By the assumption, $|f(z_0 + re^{it})| \leq |f(z_0)|$. We have

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| dt \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0)| dt = |f(z_0)|.$$

It follows that $|f(z_0 + re^{it})| = |f(z_0)|$ for all $t \in [0, 2\pi]$ and all $0 < r < R$. Thus, $|f|$ is constant in $D(z_0; R)$. By part (a), f is constant in $D(z_0; R)$.

- (c) The set $\overline{G}$ is bounded and closed and thus compact, and the continuous function f attains its maximum at some point $z_0 \in \overline{G}$.

Assume that $z_0 \in G$. Since G is open, there is $R > 0$ with $D(z_0; R) \subseteq G$. We have $|f(z_0)| \geq |f(z)|$ for all $z \in D(z_0; R)$. Then by part (b) f is constant in $D(z_0; R)$. By the Identity Theorem, f is constant in G .

We conclude that a non-constant function cannot attain its maximum in G .